

Age, symptoms, and functional outcome after mild traumatic brain injury

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Funding information

The study was financially supported by the Finnish State Research Funding and the Finnish Medical Society Duodecim.

Objectives: Previous older adult studies have examined outcome from moderate-to-severe traumatic brain injury (TBI), but fewer have focused on the effects of milder brain injuries. This study examined age-related differences in symptom reporting and whether symptoms are differentially related to functional outcome based on age.

Materials & Methods: Patients presenting with a head injury at the Tampere University Hospital emergency department (N = 325, aged 18-100) were evaluated at 1 week post injury on the Modified Rankin Scale and Rivermead Post Concussion Symptoms Questionnaire (RPQ).

Results: A quarter of participants had missing RPQ data, with missingness associated with greater age and strongly associated with pre-existing functional impairment, dementia, and/or neurological conditions. Among participants with complete data, participants <65 years old endorsed headaches, noise/light sensitivity, irritability, and frustration/impatience at a greater frequency than participants ≥65 years old. However, no differences were found in the number of symptoms endorsed or the total symptom severity score. The correlations between the severity of symptoms and change in function were similar between the two age groups.

Conclusions: Older adults tended to report fewer symptoms, but symptoms had a roughly equivalent relationship with declines in functioning across age groups. A large percentage of older adults in this study had pre-injury dementia or neurological disease that contributed to missing data on the 1-week outcome measures. The results provide insight into the impact of mild spectrum TBI on older adults compared to younger patients.

KEYWORDS

aged, brain concussion, head injuries: closed, outcome assessment (health care), post-concussion syndrome

1 | INTRODUCTION

A nationwide study spanning 15 years (between 1991 and 2005) discovered that the incidence of traumatic brain injury (TBI) requiring hospitalization in patients 70 years or older increased by

59.4%.¹ This indicates a need to better understand natural recovery, prognosis, and rehabilitation needs of older adults following TBI. Although clinical and research interest in TBI in older adults has grown,² mild TBI (MTBI) in this age group has garnered less scientific attention. An abundance of literature has examined MTBI among younger samples,³ but far fewer studies have explored MTBI

outcomes among older adults.⁴⁻¹² MTBI incidence has different characteristics among older adults compared to younger samples. Whereas youth tends to suffer MTBI due to sport-related injuries or motor vehicle accidents, MTBI in older adults results primarily from falls,¹³ which are related to frailty,¹⁴ cognitive impairment,¹⁵ and dementia severity.¹⁶ To date, some findings have indicated poor neuropsychological⁶ and functional outcomes⁴ from MTBI among older adults; however, studies using orthopedic controls have found no difference in performance attributable to the presence of an MTBI on neuropsychological outcomes.^{7,8} Despite good neuropsychological recovery, older adults patients with MTBI may continue to report anxiety, depression, and somatic symptoms 1-2 months post-injury.¹⁷ At emergency evaluation, older adults seem to present with higher Glasgow Coma Scale (GCS) than younger patients with similar anatomic TBI severity.^{18,19} This is problematic because post-injury GCS scores are widely used to stratify TBI severity, especially in clinical studies; and studies on MTBI frequently examine older adults together with younger patients. Older adults with more severe TBIs can be misclassified as mild cases, although anatomically these patients rightfully represent moderate TBI with predominantly unfavorable outcome.

Some previous research has examined differences in MTBI outcomes based on age, comparing younger samples to older adult samples on varied outcomes. A pair of studies on functional outcomes examined patients with mild spectrum TBI that were hospitalized following injury, finding older adults were more likely to be discharged to a rehabilitation setting⁹ and have longer hospital stays with worse functional, cognitive, and physical outcomes.¹⁰ Another pair of studies on outpatients with MTBI^{11,12} examined age-related differences in outcomes following injury, finding *better* outcome among older age groups. Comparing patients ≥ 60 years old with those 18-59 years old within, on average, the first month post injury, older patients had better functional, psychosocial, psychological, and post-concussion symptom outcomes. A more recent study¹¹ examined post-concussion symptoms around a year after MTBI, finding that older adults (ie, ≥ 66 years old) reported a lower severity of symptoms (eg, headache, nausea/vomiting, irritability, poor concentration, and slowed thinking) compared to middle-aged patients (ie, 36-45 years old).

The relationship between age and outcome following MTBI is not well understood and notably understudied.² Research on participants with mild spectrum TBI hospitalized after injury show worse outcome among older adults,^{9,10} whereas those studies on outpatients with MTBI show better outcome.^{11,12} An improved understanding of outcome following MTBI in older adults would be valuable for clinicians to better determine prognosis and identify rehabilitation needs. The purpose of this study is to determine if there are age-related differences in symptom reporting and if symptoms are differentially related to functional outcome. Although the literature is mixed, we hypothesized that older adults would report fewer post-concussion symptoms and evidence a weaker association between those symptoms and their functional outcome following mild spectrum TBI.

2 | MATERIALS AND METHODS

2.1 | Participants

This study was approved by the Ethical Committee of Pirkanmaa Hospital District, Tampere, Finland (ethical code: R15045). All participants provided written informed consent according to the Declaration of Helsinki. Participants included adult patients between the ages 18 and 100 ($n = 325$) evaluated for head injury upon presentation to the Tampere University Hospital Emergency Department (ED) between November 2015 and November 2016. All participants presented to the ED, and none were recruited from other clinics. Exclusion criteria included GCS < 13 ($n = 5$), undergoing neurosurgery or another type of surgery due to the acute injury ($n = 7$), having a second head injury within a week of the first injury ($n = 2$), or dying within a week of the injury ($n = 2$). This resulted in a sample size of 309. Research data are not shared.

Among this sample, 88 participants had missing data on the Rivermead Post Concussion Symptoms Questionnaire (RPQ) at 1 week post injury, which was the primary outcome of interest. These individuals were also excluded, resulting in the final sample of 221 participants included in the analyses ($M_{\text{Age}} = 56.61$, $SD = 21.82$, range: 18-93; 46.6% men). This sample was divided into participants below 65 years old ($n = 120$; $M_{\text{Age}} = 36.63$, $SD = 13.96$, range: 18-64; 52.5% men) and those 65 years old or greater ($n = 101$; $M_{\text{Age}} = 76.77$, $SD = 7.72$, range: 65-93; 39.6% men). For participants below 65 years old, mechanisms of injury included falls ($n = 76$; 63.3%), sport ($n = 13$; 10.8%), violence ($n = 10$; 8.3%), bicycle accidents ($n = 6$; 5%), motor vehicle accidents ($n = 5$; 4.2%), and other/unknown mechanisms ($n = 10$; 8.3%). For participants 65 years old and above, the mechanisms of injury included falls ($n = 96$; 95.0%), sport ($n = 1$; 1.0%), violence ($n = 1$; 1.0%), and other/unknown mechanisms ($n = 3$; 3.0%).

The total sample can be characterized as a *mild spectrum TBI* group, inclusive of all participants who presented to the ED and were evaluated for MTBI by a physician. These participants had minimal, mild, or moderate head injuries based on categorizations used by the Scandinavian guidelines,²⁰ with a single participant categorized as moderate based on a GCS of 13 (which would be considered mild in severity based on other established definitions of TBI).^{21,22} There was substantial missing information in the medical records regarding classic injury severity indicators and other signs of injury suggesting a probable MTBI. Of the 221 participants, 141 (63.8%) were referred for a CT scan of their head and brain due to concern about possible trauma-related intracranial abnormalities. More than half of the participants ($n = 123$; 55.7%) had at least one of the following clinical signs documented by the on-call physician: loss of consciousness (LOC) ($n = 72$; 32.6%), post-traumatic amnesia (PTA) ($n = 80$; 36.2%), post-traumatic seizure ($n = 4$; 1.8%), GCS 13-14 ($n = 8$, 3.6%), focal neurological deficit ($n = 17$; 7.7%), skull fracture ($n = 4$; 1.8%), or intracranial abnormality on computed tomography (CT) ($n = 16$; 7.2%). LOC was considered present if eye-witnessed or if LOC was suspected and the ED physician referred the participant for CT. Of the 221 participants included in the analyses, 80 (36.2%) were not

referred for head CT, and 56 (25.3%) of those did not have any of the above injury signs clearly documented by the on-call physician. Of note, among the 141 participants referred for a head CT to rule out intracranial bleeding, 42 (29.8%) had no clinical signs of TBI documented, indicating that the absence of documented signs does not mean that an MTBI was not suspected. These participants were retained because they presented to the ED, were evaluated for a head injury, and qualified for at least a minimal head injury under Scandinavian guidelines.²⁰

2.2 | Procedure

The on-call ED physician rated the GCS of each participant and collected data on additional TBI signs. The prehospital ambulance personnel documents were utilized to collect injury-related data (eg, presence of LOC). CT referrals were based on Scandinavian guidelines,²³ with scans coded by a radiologist according to National Institute of Neurological Disorders and Stroke Common Data Elements.²⁴ The medical history of each participant was available through their electronic record, which included a local hospital and all healthcare centers in Tampere, Finland. Each record was reviewed for pre-existing conditions, which were coded as present or absent based on the International Classification of Diseases, Tenth Edition (ICD-10).²⁵ For the purposes of this study, relevant categories of pre-existing conditions included Dementia (ie, Alzheimer's disease [G30], vascular dementia [F01], and unspecified dementia [F03]), mental and behavioral disorders (ie, F00-F99), and diseases of the nervous system (ie, G00-99). Medical history information and injury-related data were collected retrospectively from available records, and outcome measures, described below, were collected prospectively post injury.

2.3 | Measures

A nurse with neurological training administered the Finnish-language versions of the Modified Rankin Scale (MRS) and RPQ by phone at approximately 1 week post injury. Both the nurse and participant were not blinded to CT findings prior to the administration of both measures. The MRS²⁶⁻²⁸ rates the severity of patient disability on the following scale: no symptoms (0), symptoms/no disability (1), slight disability, meaning despite symptoms, able to carry out all usual activities (2), moderate disability (3), moderate-severe disability (4), severe disability (5), and dead (6). Participants were rated on pre-injury disability severity based on retrospective self-report and post-injury disability severity based on self-report of their functioning at 1 week post injury. The Δ MRS was calculated as the difference between pre-injury and post-injury ratings, with the Δ MRS score used in all analyses. A higher score was indicative of an increase in functional impairment and disability. The RPQ²⁹ is a 16-item self-report questionnaire on post-concussion symptom severity. Each item is rated on the following 5-point scale, with the respective score for each

severity rating specified in parentheses: not experienced at all (0), no more of a problem (0), a mild problem (2), a moderate problem (3), and a severe problem (4). The coding of "no more of a problem" as 0 is consistent with the scoring procedure from the original development of the RPQ.²⁹ The current study dichotomized the RPQ symptoms into present (ie, 2-4) or absent (ie, 0) for analysis. The total number of symptoms endorsed (range: 0-16) and the RPQ total severity score were also used. The RPQ total score is based on the summing of all item severity ratings (ie, 0-4 for each item). This score ranges from 0 to 64, with a higher score indicative of greater symptom severity.

2.4 | Statistical analysis

Significance was set at $P < .05$. Due to the exploratory nature of the study, there was no a priori correction made for multiple comparisons. Consistent with previous research on age-related differences in TBI,^{18,19} participants were grouped as <65 years old and ≥ 65 years old. The frequencies of symptom endorsement were compared across age groups for each RPQ symptom using *chi-squared tests*. Spearman ρ correlations were calculated between Δ MRS and the RPQ total score and three composite scores: cognitive, emotional, and somatic. These three composite scores for the RPQ were based on previous empirical research supporting a three-factor RPQ model³⁰ and the replication of this model using a sample with MTBI from northern European EDs.³¹

A follow-up analysis dichotomized MRS values into ratings of no functional impairment (MRS = 0-1) and functional impairment (MRS = 2-5; note that no participants had post-injury ratings of 6, which correspond to death) for both retrospective and post-injury scores. One group included participants that transitioned from having no functional impairment prior to their injury (per retrospective MRS rating) to having functional impairment after their injury ($n = 122$). Another group included participants that did not change in their functional impairment status ($n = 99$), meaning they either remained unimpaired (ie, MRS remained 0 or 1) or had pre-existing impairment that worsened (ie, MRS of 2 or greater increased after injury). Using a Mann-Whitney U test, the RPQ total symptom severity score was compared between younger and older adult participants for both those who transitioned to functional impairment after their injury (ie, went from MRS = 0-1 to MRS = 2-5) and those who had no change in their functional impairment status.

3 | RESULTS

3.1 | Evaluation of missingness on RPQ

A large portion of the original sample was missing the RPQ. Numerous variables were associated with missingness (Table 1). Individuals with missing RPQ were older and more likely men. They were much more likely to have pre-existing functional impairment,

	Valid RPQ (n = 221)	Missing RPQ (n = 88)	$U/\chi^2, P$
Demographics			
Age (Mdn, IQR)	60, 38-75	72, 55-85.5	$U = 12\,502$, $P < .001$
≥65 years old (%)	45.7	62.5	$\chi^2 = 7.11$, $P = .008$
Men (%)	46.6	60.2	$\chi^2 = 4.67$, $P = .031$
Pre-injury functioning and conditions			
Pre-injury MRS (Mdn, IQR)	0, 0-2	3, 2-3	$U = 10\,281.5$, $P < .001$
Pre-injury disability (MRS 2-5, %)	39.4	79.3	$\chi^2 = 29.39$, $P < .001$
Pre-injury moderate disability (MRS 3-5, %)	10.4	63.8	$\chi^2 = 77.57$, $P < .001$
Pre-injury disability (MRS 2-5) or dementia (%)	40.7	80.0	$\chi^2 = 29.13$, $P < .001$
Pre-injury moderate disability (MRS 3-5) or dementia (%)	13.6	66.1	$\chi^2 = 69.19$, $P < .001$
Dementia (%)	4.5	25.3	$\chi^2 = 28.90$, $P < .001$
Diseases of the nervous system (%)	32.6	45.5	$\chi^2 = 4.52$, $P = .034$
Mental and behavioral disorders (%)	43.9	52.3	$\chi^2 = 1.78$, $P = .182$
Mental, behavioral, or neurological conditions (%)	61.5	76.1	$\chi^2 = 5.95$, $P = .015$
Injury characteristics			
GCS < 15 (%)	3.6	4.5	$\chi^2 = 0.144$, $P = .704$
LOC (%)	32.6	34.1	$\chi^2 = 0.065$, $P = .799$
PTA (%)	36.2	35.2	$\chi^2 = 0.026$, $P = .872$
Positive CT (%)	11.3	13.2	$\chi^2 = 0.155$, $P = .694$

Note: LOC was defined as present if either eye-witnessed, or if suspected with subsequent referral to CT.

Abbreviations: CT, computed tomography; GCS, Glasgow Coma Scale; LOC, loss of consciousness; MRS, Modified Rankin Scale; PTA, post-traumatic amnesia; RPQ, Rivermead Post Concussion Symptoms Questionnaire.

TABLE 1 Differences in demographic, pre-injury health and functioning, and injury characteristics related to missingness on the RPQ

dementia, and neurological conditions. Missingness was unrelated to injury characteristics, including frequencies of GCS < 15, LOC, PTA, or positive CT.

3.2 | Age group differences

Differences were also evident between age groups on multiple variables (Table 2). The older age group (ie, ≥65 years old) was much more likely to have pre-existing functional impairment or dementia than the younger age group. Participants under 65 years old had a greater frequency of PTA than participants 65 years and older, but the age groups did not differ in the frequency of any other injury characteristic. For both age groups, there were no significant differences in the total number of symptoms endorsed (ie, <65 years old: $U = 1653$, $P = .588$; ≥65 years old: $U = 1252.5$, $P = .889$) or total symptom severity (ie, <65 years old: $U = 1649.5$, $P = .575$; ≥65 years old: $U = 1245$, $P = .847$) at 1 week post injury on the RPQ in those that had documented clinical signs of MTBI vs those who did not.

The younger age group endorsed the following symptoms at a significantly greater frequency: headaches, noise sensitivity, irritability, frustration/impatience, and light sensitivity (see Table 3 for exact rates of symptom endorsement). Older adults endorsed no symptoms at a significantly greater frequency than younger adults. There were no significant group differences in the number of symptoms endorsed or the total severity score for the RPQ.

3.3 | Functional status and composite symptom scores

For participants under 65 years old, the correlation between Δ MRS and the RPQ total score was $\rho = .601$, $P < .001$. The correlations between Δ MRS and the RPQ composite scores were as follows: cognitive symptoms, $\rho = .354$, $P < .001$; emotional symptoms, $\rho = .342$, $P < .001$; and somatic symptoms, $\rho = .589$, $P < .001$. For participants 65 years old or older, the correlation between Δ MRS and the RPQ total score was $\rho = .570$, $P < .001$. The correlations between Δ MRS

TABLE 2 Age group differences on demographic, pre-injury health and functioning, and injury characteristics

	<65 years old (n = 120)	≥65 years old (n = 101)	U/ χ^2 , P
Demographics			
Age (Mdn, IQR)	41.0, 26-51	76, 71-83	U = 12 120, P < .001
Male (%)	52.5	39.6	$\chi^2 = 3.67$, P = .056
Pre-injury health and functioning			
Pre-injury MRS (Mdn, IQR)	0, 0-2	2, 0-2	U = 8485, P < .001
Pre-injury disability (MRS 2-5, %)	26.7	54.5	$\chi^2 = 17.74$, P < .001
Pre-injury moderate disability (MRS 3-5, %)	2.5	19.8	$\chi^2 = 17.61$, P < .001
Pre-injury disability (MRS 2-5) or dementia (%)	26.7	57.4	$\chi^2 = 21.50$, P < .001
Pre-injury moderate disability (MRS 3-5) or dementia (%)	2.5	26.7	$\chi^2 = 27.45$, P < .001
Dementia (%)	0	9.9	P < .001 ^a
Diseases of the nervous system (%)	27.5	38.6	$\chi^2 = 3.08$, P = .079
Mental and behavioral disorders (%)	46.7	40.6	$\chi^2 = 0.821$, P = .821
Mental, behavioral, or neurological conditions (%)	61.7	61.4	$\chi^2 = 0.002$, P = .966
Injury characteristics			
GCS < 15 (%)	5.0	2.0	P = .295 ^a
LOC (%)	35.8	28.7	$\chi^2 = 1.27$, P = .261
PTA (%)	43.3	27.7	$\chi^2 = 5.79$, P = .016
Positive CT (%)	9.4	13.0	$\chi^2 = 0.453$, P = .501

Note: LOC was defined as present if either eye-witnessed, or if suspected with subsequent referral to CT.

Abbreviations: CT, computed tomography; GCS, Glasgow Coma Scale; LOC, loss of consciousness; MRS, Modified Rankin Scale; PTA, post-traumatic amnesia; RPQ, Rivermead Post Concussion Symptoms Questionnaire.

^aDesignates 2-tailed P-value for Fisher's exact test.

and the RPQ composite scores were as follows: cognitive symptoms, $\rho = .433$, $P < .001$; emotional symptoms, $\rho = .393$, $P < .001$; and somatic symptoms, $\rho = .537$, $P < .001$.

The relationship between functional status and symptom burden was further examined by selecting only participants who transitioned to some degree of functional impairment following injury (ie, moved from a retrospective pre-injury MRS rating of 0-1 to a post-injury MRS rating of 2-5). No participants transitioned from an MRS of 2-5 to an MRS of 0-1. Among participants 65 years old or older, 66 (65.3%) transitioned from no functional impairment to functional impairment, and among participants under 65 years old, 56 (46.7%) transitioned from no functional impairment to functional impairment at 1 week post injury. Among these 122 participants transitioning to functional impairment, the younger group had a higher symptom severity score on the RPQ (Mdn = 7, IQR = 1-16) than the older group (Mdn = 2, IQR = 0-6) based on a non-parametric comparison ($U = 1274$, $P < .001$). There was no difference in RPQ scores among younger (Mdn = 0, IQR = 0-2) and older participants

(Mdn = 0, IQR = 0-2) who had no change in functional impairment status at 1 week post injury ($U = 1060$, $P = .612$).

4 | DISCUSSION

There are few studies that directly compare outcome from MTBI in younger adults vs older adults. Patients presenting to the ED for evaluation after a head injury, the vast majority of whom presumably had mild spectrum TBI, were evaluated for post-concussion symptoms and changes in functioning at 1 week post injury. Younger (ie, <65 years old) and older patients (ie, ≥65 years old) were compared on the frequency of post-concussion symptom endorsement, with a few symptoms endorsed significantly more often among younger participants (ie, headache, noise and light sensitivity, irritability, and reduced frustration tolerance). Although the age groups did not differ in total symptom severity or the number of symptoms endorsed, younger participants endorsed all symptoms, except for

	<65 years old (n = 120)	≥65 years old (n = 101)	$U/\chi^2, P$
Headaches	35.0%	20.8%	$\chi^2 = 5.43, P = .020$
Feelings of dizziness	25.8%	19.8%	$\chi^2 = 1.12, P = .289$
Nausea and/or vomiting	7.5%	6.9%	$\chi^2 = 0.026,$ $P = .871$
Noise sensitivity	13.3%	4.0%	$\chi^2 = 5.85, P = .016$
Sleep disturbance	16.7%	17.8%	$\chi^2 = 0.051,$ $P = .821$
Fatigue, tiring more easily	35.8%	33.7%	$\chi^2 = 0.114,$ $P = .736$
Being irritable, easily angered	15.0%	3.0%	$\chi^2 = 9.23, P = .002$
Feeling depressed or tearful	11.7%	5.9%	$\chi^2 = 2.19, P = .139$
Feeling frustrated or impatient	13.3%	4.0%	$\chi^2 = 5.85, P = .016$
Forgetfulness, poor memory	17.5%	10.9%	$\chi^2 = 1.93, P = .164$
Poor concentration	17.5%	8.9%	$\chi^2 = 3.45, P = .063$
Taking longer to think	15.0%	7.9%	$\chi^2 = 2.65, P = .104$
Blurred vision	11.7%	8.9%	$\chi^2 = 0.447,$ $P = .504$
Light sensitivity	15.0%	4.0%	$\chi^2 = 7.46, P = .006$
Double vision	1.7%	0.0%	$P = .501^a$
Restlessness	9.2%	4.0%	$\chi^2 = 2.35, P = .125$
RPQ—Number of symptoms endorsed (Mdn, IQR)	1, 0-4	1, 0-2	$U = 5492, P = .208$
RPQ—Total score (Mdn, IQR)	2, 0-9	2, 0-5	$U = 5465, P = .186$

Note: Bold values indicate significance at $P < .05$.

Abbreviation: RPQ, Rivermead Post Concussion Symptoms Questionnaire.

^aDesignates 2-tailed P -value for Fisher's exact test.

TABLE 3 Differences in symptom endorsement frequencies by age group: participants with mild spectrum traumatic brain injury (N = 221)

sleep disturbance, at a greater rate than participants aged 65 years and older. Despite the trend toward greater symptom endorsement among younger participants, the correlation between the severity of symptoms and change in functional status was similar between the two age groups, indicating that, even with lower rates of symptom endorsement, the relationship between symptoms and reductions in daily functioning was similar. Younger participants who transitioned to having functional impairment post injury reported significantly more symptoms than older adults who transitioned to having functional impairment, albeit many older adults had pre-injury functional impairment based on their retrospective MRS ranking. This finding may indicate that, although older adults tend to report fewer symptoms, these symptoms tend to have a roughly equivalent relationship with declines in functioning. These findings appear consistent with previous findings indicating better symptom outcomes following MTBI among older adults compared to younger samples.^{11,12}

Previous researchers examining age-related differences in symptom reporting among older adults have attributed these differences to (a) minimal perceived change in symptoms compared to their premorbid status among older adults or (b) limited co-occurring employment-related and life stressors among older adults.¹¹ Another study found older adults reported better global functioning and less psychosocial, psychological, and physical impairment than younger

patients; however, after controlling for employment status, no significant differences were observed between the age groups.¹²

The results of this study may have been influenced, at least to a small degree, by missing data on the primary outcome of interest in a larger subgroup of older adults compared to younger adults. Older adults and those with mild vs moderate-to-severe TBI tend to seek medical treatment less often,³² and those with significant levels of pre-existing impairment may be less likely to participate in outcome research, as demonstrated in the current study by variables associated with missingness on the self-report RPQ (eg, pre-existing dementia or neurological conditions). The substantial missingness of RPQ data is likely attributable to pre-existing conditions resulting in functional or cognitive impairments that could impact the capacity of patients to actively participate in research or complete a self-report questionnaire. Previous studies that have examined psychological and symptom-related outcomes from MTBI among older adults via self-report may be based on a healthier sample of participants than the total population of those presenting to the ED following head trauma. Future studies could employ informant reports of post-concussion symptoms completed by caregivers, peers, or family members, which would allow for the inclusion of patients with more severe premorbid impairments who are unable to respond to questionnaires independently.

Aside from missing data, this study had additional limitations that may have influenced the findings. Data were collected at 1 week after presentation to the ED, so we could not examine age differences in the sequelae of a mild spectrum TBI beyond the acute post-injury phase. Both the MRS and RPQ were administered by phone, and overall, there was low symptom endorsement across age groups, with 44.2% of younger participants and 45.5% of older participants reporting no change in symptoms at 1 week following injury. Prior studies using the RPQ following MTBI have shown greater levels of symptom endorsement. One British study³³ on ED patients with head injury evaluated 3-6 months post-injury found that only 17.3% of participants reported no symptoms. Another study from the United Kingdom³⁰ on patients with primarily MTBI presenting to the ED or trauma services found that 42% of participants endorsed no symptoms at 6 months post-injury. These studies illustrate that a substantial percentage of our sample appeared mostly or entirely clinically recovered by 1 week post-injury, which could result in part from the enrollment of patients who had very mild injuries. Moreover, some studies with longer follow-up intervals might have somewhat inflated rates of ongoing symptom endorsement in part due to greater attrition of participants who are doing well clinically and thus being more likely to drop out of the study. At least one past study on ED patients with MTBI³⁴ found that participants retained at follow-up assessments tended to have injuries of greater severity, in that attrition was related to shorter durations of LOC, PTA, and retrograde amnesia.

Differences in symptom endorsement frequency may also result from the phone administration of the RPQ. Previous studies have shown greater symptom endorsement on self-report questionnaires than during an interview.^{35,36} There might also be cultural differences in symptom endorsement, where Scandinavian patients might report fewer symptoms compared to patients from other geographical regions. A study examining symptom reporting at 3 months post-injury among Swedish ED patients presenting with MTBI³¹ found that 56% of participants were at the floor of the RPQ. The frequency of these participants endorsing an absence of symptoms was higher than the current study, but data were also collected more distal from the injury event.

Study setting should also be considered when interpreting and drawing inference from the findings. The sample represents a cohort of sequentially recruited patients who presented to a single ED in Finland following a mild head injury. Due to the clinical environment, documentation within the ED is often terse, and prehospital paramedic records can be incomplete or include insufficient detail about clinical signs of brain injury. The on-call physician did not document clinical signs of brain injury in a manner that allowed for a retrospective diagnosis of MTBI. LOC and PTA durations were not recorded and may exceed durations that qualify as mild spectrum TBI in some cases. Further, many participants had a GCS of 15 without documented LOC or PTA, which may correspond to a mild head injury that does not meet clinical criteria for a MTBI.

Despite limitations, the examination of post-concussion symptoms among older adults stands as an important area of research. A

growing interest in TBI in older adults has focused largely on moderate-to-severe injuries, and few researchers have examined outcomes from the milder spectrum of TBI in this age group. Currently, there are few to no evidence-based guidelines for acute inpatient and long-term outpatient care for older adults with any severity of TBI, both nationally or internationally,² and research on prognosis and interventions among older adults is warranted. This study provides additional information on symptom and functional outcome following mild spectrum TBI among older adults in the acute phase of recovery.

ACKNOWLEDGMENT

The authors acknowledge research assistant Anne Simi for her assistance with the patient enrollment and data collection at Tampere University Hospital.

CONFLICT OF INTEREST

Teemu Luoto, MD, PhD, has received funding from the Government's Special Financial Transfer tied to academic research in Health Sciences (Finland). Grant Iverson, PhD, has a clinical and consulting practice in forensic neuropsychology involving individuals who have sustained mild TBIs. He has received research funding from several test publishing companies, including IMPACT Applications, Inc, CNS Vital Signs, and Psychological Assessment Resources (PAR, Inc). He has received salary support from the National Football League and the Harvard Integrated Program to Protect and Improve the Health of National Football League Players Association Members. He serves as a scientific advisor for Sway Operations, LLC, Highmark, Inc, and BioDirection, Inc. He acknowledges unrestricted philanthropic support from IMPACT Applications, Inc, the Heinz Family Foundation, and the Mooney-Reed Charitable Foundation.

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How to cite this article: Karr JE, Luoto TM, Gilman IG, Berghem K, Kotilainen A-K, Iverson GL. Age, symptoms, and functional outcome after mild traumatic brain injury. *Acta Neurol Scand.* 2020;141:183-190. <https://doi.org/10.1111/ane.13190>